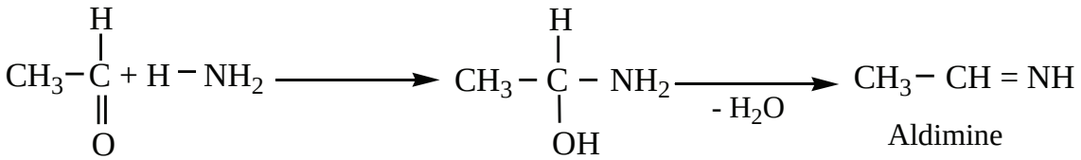
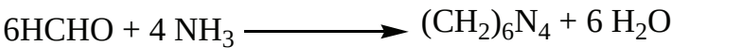
5) Reaction with NH_3

i) Aliphatic aldehydes (except formaldehyde) react with an ethereal solution of ammonia to form aldimines.



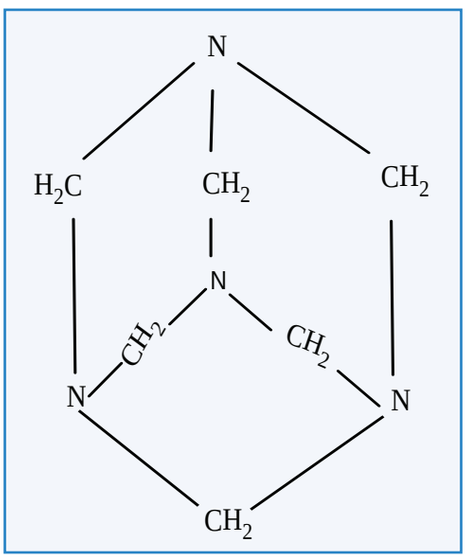
Acetaldehyde — Acetaldehyde ammonia — Aldimine

ii) Formaldehyde reacts with ammonia to form hexa methylene tetramine, which is also known as **Urotropine**.



Formaldehyde — Hexamethylene tetramine

Structure

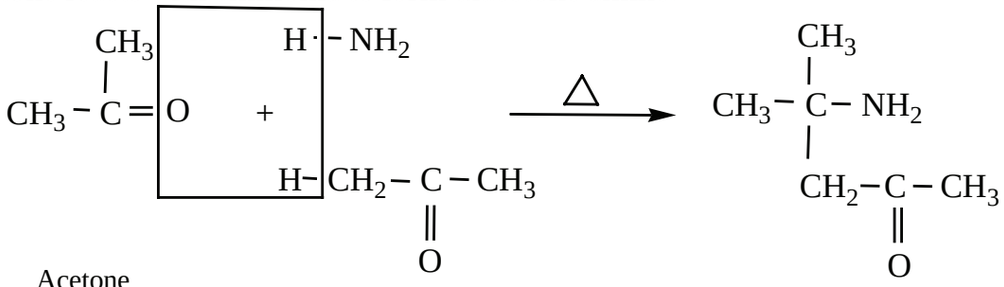


Uses

(i) Urotropine is used as a medicine to treat urinary infection.

(ii) Nitration of Urotropine under controlled condition gives an explosive RDX (Research and development explosive). It is also called cyclonite or cyclotri methylene trinitramine.

iii) Acetone reacts with ammonia to form diacetone amine.



Acetone — Acetone — Diacetone amine

XII U12-Carbonyl Compounds.indd 158 11-01-2021 16:10:37